Keep settings where:

$$10^{-4} \lesssim \delta_{\text{avg}} \lesssim 10^{-1},$$

and where p_H varies noticeably across keys.

Output:

`threshold_scan.csv`

Columns:

```
r,d,t,I,threshold_schedule,num_keys,num_errors_per_key,
avg_pH,min_pH,median_pH,max_pH,num_zero_failure_keys
```

This experiment is not a final result; it selects useful stress parameters.

Experiment 1 — Exact toy full-domain certification

Goal: prove the framework on a fully enumerable finite domain.

Use very small parameters, for example:

$$(r, d, t) \in \{(13, 3, 2), (13, 3, 3), (17, 3, 3), (19, 3, 3)\}.$$

For each parameter set:

1. Enumerate all keys, or enumerate modulo cyclic symmetry if full enumeration is too large.
2. Enumerate all errors $e \in E_t$.
3. Run the decoder.
4. Compute exact:

$$p_H = \frac{1}{|E_t|} \sum_{e \in E_t} 1[\mathrm{Dec}_H(s_H(e)) \neq e].$$

5. Compute exact:

$$\delta_{\text{avg}}, \quad \Pr_H[p_H > \tau], \quad \mathrm{Fail}_q = \mathbb{E}_H[1 - (1 - p_H)^q].$$

The attached paper defines the exact reduced-parameter DFR in this form and says exact enumeration is the certificate-grade case; sampled estimates are diagnostics unless converted into one-sided bounds. dependency_aware_dfr_bounds_ste...

Output:

`exact_toy_key_results.csv`

```
param_id,key_id,r,d,t,threshold_schedule,pH,
num_errors,num_failures,num_succ,num_timeout,
num_wrongzero,num_nonzerohalt
```

`exact_toy_tail_curves.csv`

```
param_id,tau,exact_tail_prob,delta_avg,
q,exact_fail_q,union_q_delta_avg
```

Required figure:

Exact tail curve:

$$\tau \mapsto \Pr_H[p_H > \tau].$$

This is the anchor experiment. It gives the paper one theorem-grade finite domain.

Experiment 2 — Profile extraction experiment

Goal: compute the structural profile $\Phi(H)$ for every certified key.

For each key in Experiments 1 and later experiments, compute:

$$D_{ab}(\Delta), \quad C_4^{\mathrm{loc}}(H), \quad D_{\max}(H), \quad \Lambda_H,$$

near-codeword overlap tails:

$$A_H^{\mathcal{N}}(u),$$

low-syndrome gathering counts:

$$G_H(m, \ell),$$

and fixed-point / absorbing indicators:

$$B_H(M, T).$$

The paper's profile coordinates are exactly these: cyclic-overlap spectra, local cycle pressure, near-codeword overlap tails, gathering indicators, and fixed-point indicators.

dependency_aware_dfr_bounds_ste...

Output:

`profiles.csv`

```
param_id,key_id,
C4_loc,Dmax,Lambda_json,
near_tail_json,gathering_json,fixedpoint_json,
profile_M1_bin,profile_M2_bin,profile_M3_bin
```

Where:

- **M1** uses coarse decoder-state information only.
- **M2** adds near-codeword overlap.
- **M3** uses the full proposed profile.

Experiment 3 — Direct profile-bin bound experiment

Goal: test whether the profile separates high-p_H keys.

For each diagnostic model:

M0 — average-only / independence-only

No meaningful key structure.

M1 — syndrome-weight-only

Profile uses only coarse quantities such as:

$$|E|, \quad |s_H(E)|.$$

M2 — near-codeword-only

Profile uses:

$$|E|, \quad |s_H(E)|, \quad U_H^{\mathcal{N}}(E).$$

M3 — full dependency-aware profile

Profile uses:

$$C_4^{\mathrm{loc}}, D_{\max}, \Lambda_H, A_H^{\mathcal{N}}, G_H, B_H.$$

For each profile bin C_φ, compute:

$$p_\varphi^{\max} = \max_{H \in C_\varphi} p_H.$$

Then compute profile-tail bounds:

$$\Pr[p_H > \tau] \le \sum_{\varphi : p_\varphi^{\max} > \tau} \nu_\varphi.$$

Also compute average bounds:

$$\delta_{\text{profile}} = \sum_\varphi \nu_\varphi p_\varphi^{\max}.$$

And repeated-use bounds:

$$\text{Fail}_q \le \sum_\varphi \nu_\varphi \left(1 - (1 - p_\varphi^{\max})^q\right).$$

Output:

`profile_bin_bounds.csv`

```
param_id,model,bin_id,num_keys,bin_mass,
p_bin_max,p_bin_mean,p_bin_median,
delta_contribution
```

`profile_tail_bounds.csv`

```
param_id,model,tau,exact_tail,bound_tail,
tail_gap,tail_ratio
```

Success criterion:

M3 should produce tighter tail bounds than M0, M1, and M2.

This experiment directly tests whether the full profile is useful.

Experiment 4 — Full finite-state upper-process certificate

Goal: show that the actual proposed certificate is non-vacuous.

For every enumerated decoder trajectory, log:

$$E_i, \quad s_i, \quad F_i, \quad M_i = E_i \setminus F_i, \quad L_i = F_i \setminus E_i, \quad \chi_H(E_i).$$

The paper says this trajectory record directly instantiates the count-envelope construction and avoids independence assumptions between bit decisions.

dependency_aware_dfr_bounds_ste...

For each model M1/M2/M3, profile bin φ, iteration i, and state x, compute:

$$Q_i^+(\varphi, x; m, \ell) = \max_{H \in C_\varphi} \Pr[|M_i| = m,\ |L_i| = \ell \mid \chi_H(E_i) = x].$$

Then propagate the non-normalized finite-state upper process and obtain:

$$p^+_{\varphi,\mathrm{FS}}.$$

Compare it against:

$$p^{\max}_\varphi.$$

Output:

`trajectory_logs.parquet`

```
param_id,key_id,error_id,iteration,
state_id,residual_weight,syndrome_weight,
near_overlap_bin,fixedpoint_bin,omega_bin,
num_missed,num_false_flips,
terminal_status
```

`finite_state_bounds.csv`

```
param_id,model,bin_id,
p_bin_exact_max,p_fs_upper,
looseness_ratio,additive_gap,
num_states,num_transitions,row_sum_max
```

Success criterion:

For all keys:

$$p_H \le p^+_{\kappa(H),\mathrm{FS}}.$$

Also, the bound should be non-vacuous:

- most $p^+_{\varphi,\mathrm{FS}} < 1$,
- median looseness not enormous,
- M3 finite-state bounds tighter than M1/M2.

Experiment 5 — Baseline comparison and ablation

Goal: prove that the full dependency-aware profile adds information beyond simpler features.

Compare:

Model	Coordinates
M0	average / independence only
M1	syndrome weight only
M2	M1 + near-codeword overlap
M3a	M2 + cycle spectrum
M3b	M3a + gathering counts
M3c	M3b + fixed-point indicators
M3	full profile

For each model, report:

```
coverage
median looseness
95th percentile looseness
worst looseness
tail-curve error
Spearman correlation with exact pH
top-5% weak-key recall
```

The paper already lists coverage, looseness, tail accuracy, rank separation, and multi-query amplification as primary diagnostics. dependency_aware_dfr_bounds_ste...

Output:

`model_comparison.csv`

```
param_id,model,
coverage,median_looseness,p95_looseness,worst_looseness,
tail_l1_error,tail_linf_error,
spearman_pH,top1_recall,top5_recall,top10_recall,
failq_bound_q1e2,failq_bound_q1e4,failq_bound_q1e6
```

Success criterion:

M3 should clearly beat M2. If M2 performs just as well, then the paper's novelty should be narrowed to a near-codeword profile-tail framework rather than a broad dependency-aware framework.

Experiment 6 — Exact sampled reduced-parameter experiment

Goal: scale beyond tiny full-domain enumeration while keeping exact error enumeration per sampled key.

Use the paper's proposed primary grid:

$$r \in \{31, 47\}, \quad d \in \{5, 7\}, \quad t \in \{3, 4, 5\}.$$

For each parameter set:

1. Sample uniform keys.
2. Add profile-stratified keys from extreme structural regions:
   - high C_4^{loc},
   - high $D_{\max}$,
   - high gathering count G_H,
   - high fixed-point count B_H,
   - high near-codeword overlap tail.
3. Enumerate all errors when feasible.

For example:

$$\binom{62}{4} = 557{,}845$$

for $r = 31, t = 4$, which is feasible per key.

Output:

Same as Experiments 2–5, but with `experiment_layer=exact_sampled` .

Success criterion:

The same qualitative result should hold:

$$\text{M3 beats M2 beats M1/M0}$$

on tail separation and upper-bound tightness.

B. Strongly recommended extra experiments

Experiment 7 — Rare-event near-codeword conditioning

Goal: test whether high-p_H keys are explained by near-codewords alone or by interaction with cycle/gathering/fixed-point structure.

Sample errors conditional on:

$$U_H^{\mathcal{N}}(e) \geq u.$$

For each key and threshold u, estimate:

$$p_{H,u}^{\mathcal{N}} = \Pr[\mathrm{Dec}_H(s_H(e)) \neq e \mid U_H^{\mathcal{N}}(e) \geq u].$$

Then compute contribution diagnostic:

$$A_H^{\mathcal{N}}(u) \cdot p_{H,u}^{\mathcal{N}}.$$

The paper explicitly includes a rare-event layer because uniform sampling cannot directly reveal extremely small DFR values. dependency_aware_dfr_bounds_ste...

Output:

`rare_event_nearcodeword.csv`

```
param_id,key_id,u,
near_tail_mass_AHu,
conditional_trials,conditional_failures,
conditional_failure_rate,
contribution_proxy
```

Key question:

Within keys having similar near-codeword overlap, does M3 still separate high-failure keys?

If yes, this supports the dependency-aware novelty.

Experiment 8 — Held-out generalization experiment

Goal: show that profile bins are not overfitted to one sampled key set.

For sampled reduced parameters:

1. Split keys into train/test.
2. Choose binning rules using train keys only.
3. Compute profile scores on test keys.
4. Evaluate rank separation and top-tail recall on test keys.

Output:

`heldout_generalization.csv`

```
param_id,model,split_id,
train_num_keys,test_num_keys,
test_spearman,test_top5_recall,
test_tail_l1_error,test_tail_linf_error
```

Important: do not call this a certificate unless one-sided statistical envelopes are added. Call it a diagnostic generalization test.

Experiment 9 — Confidence-bound larger reduced experiment

Goal: provide a bridge toward larger instances without claiming production-parameter proof.

Use:

$$r \in \{61, 89\}$$

with selected:

$$d \in \{7, 9\}, \quad t \in \{5, 6, 7\}.$$

For each key, sample errors uniformly without replacement and compute a one-sided confidence upper bound on p_H.

Report:

$$\hat{p}_H \quad \text{and} \quad p_H^{\text{UCB}}.$$

The paper's protocol says larger selected parameters should use sampled errors with one-sided confidence bounds, and sampled estimates are diagnostics unless converted into explicit one-sided bounds. dependency_aware_dfr_bounds_ste...

Output:

`confidence_large_reduced.csv`

```
param_id,key_id,num_trials,num_failures,
p_hat,p_upper_confidence,confidence_level,
profile_bin_M1,profile_bin_M2,profile_bin_M3
```

Success criterion:

High M3-risk keys should have higher empirical or upper-confidence failure than low M3-risk keys.

Do not use this as the main proof. Use it as structural support.

Experiment 10 — Key-filter what-if experiment

Goal: evaluate whether profile-based rejection would remove high-p_H keys.

For a risk score $s(H)$, reject the top:

$$1\%, \quad 5\%, \quad 10\%$$

of keys by score.

Measure accepted-key:

$$\delta_{\text{avg}}^{\text{accepted}}, \quad \Pr[p_H > \tau \mid H \text{ accepted}], \quad \text{Fail}_q^{\text{accepted}}.$$

Output:

`filter_whatif.csv`

```
param_id,model,rejection_rate,
accepted_mass,delta_before,delta_after,
max_pH_before,max_pH_after,
tail_before,tail_after,
failq_before,failq_after
```

This is optional because the paper says filtering is an application, not the main contribution. But it can make the practical value clearer.

Experiment 11 — State-compression sensitivity

Goal: show that the finite-state process is not an artifact of one arbitrary binning.

Run M3 with different bin budgets:

$$8, \quad 16, \quad 32, \quad 64, \quad 128$$

profile bins or state bins.

Measure:

```
num_states
num_transitions
coverage
median looseness
```

```
tail error
runtime
memory
```

Output:

`state_compression_sensitivity.csv`

This tells reviewers whether the method improves smoothly as state information increases, or whether it is unstable.

Experiment 12 — Runtime and reproducibility experiment

Goal: make the artifact credible.

For every major parameter set, report:

```
key_generation_time
profile_extraction_time
error_enumeration_time
trajectory_logging_time
finite_state_bound_time
peak_memory
output_size
```

Output:

`runtime_report.csv`

The paper should state that all results are deterministic under recorded seeds and Docker commands.

Recommended final experiment matrix

I would implement the experiments in this order:

Priority	Experiment	Required for paper?	Purpose
1	Decoder-threshold calibration	Yes	Find useful stress parameters
2	Exact toy full-domain certification	Yes	Theorem-grade finite-domain proof

Priority	Experiment	Required for paper?	Purpose
3	Profile extraction	Yes	Compute $\Phi(H)$
4	Direct profile-bin bounds	Yes	Show profile separates high-p_H keys
5	Finite-state upper-process bounds	Yes	Validate actual framework
6	Baseline and ablation comparison	Yes	Prove M3 beats simpler diagnostics
7	Exact sampled reduced-parameter runs	Yes	Show scalability beyond toy domain
8	Rare-event near-codeword conditioning	Strongly yes	Show value beyond near-codeword-only
9	Held-out generalization	Recommended	Avoid overfitting criticism
10	Larger confidence-bound runs	Recommended	Conservative larger-parameter diagnostics
11	Key-filter what-if	Optional	Demonstrate application
12	State-compression sensitivity	Recommended	Show robustness
13	Runtime/reproducibility	Yes	Artifact credibility

Minimum result package for the paper

At minimum, the paper should include:

1. **Exact toy table** with exact p_H, δ_{avg}, and tail curves.
2. **Profile-tail figure** comparing exact tail vs M1/M2/M3 bounds.

3. **Finite-state certificate table** showing coverage and looseness.
4. **Ablation table** showing which coordinates improve separation.
5. **Rare-event near-codeword table** showing whether M3 adds information beyond near-codeword overlap.
6. **Exact sampled reduced-parameter table** for $r = 31$ or $r = 47$.
7. **Reproducibility table** with commands, seeds, runtime, and artifact paths.

The strongest experimental story would be:

> On fully enumerable reduced domains, M3 gives certified non-vacuous bounds on p_H, sharper profile-tail bounds than M0/M1/M2, and better high-p_H key separation. On larger sampled reduced domains, the same structural profile remains predictive under exact or confidence-bounded error sampling.

Sources